Step 7: Logging, Monitoring, and Admin Interface

7. Overview

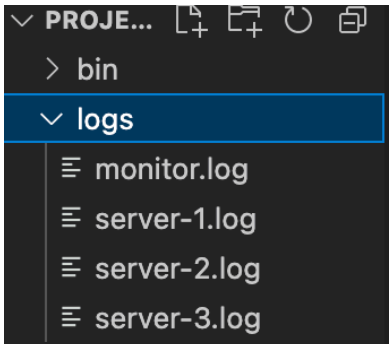
The goal of step 7 was to make the SRMS observable and manageable during execution. This would be achieved by logging in, monitoring the server health in real-time, and having an administrative interface that would allow runtime inspection and control of the system.
We were able to make SRMS observable and manageable by adding:

1. Timestamped logging for key events
2. Monitoring thorough heartbeat tracking and status reporting
3. Admin interface for health checks and manual failover actions

7.1 Logging Implementation

Each server and monitor wrote their timestamped log entries for lifecycle and runtime events. For this part, logging was implemented through a centralized logging utility that prepends timestamps and component identifiers to every log entry. Our logged events included:

- Server startup and shutdown
- Heartbeat messages sent by servers
- Heartbeat messages received by the monitor
- Failover detection and primary selection
- Promotion of a backup server to primary
- Administrative commands issued through the monitor



- *monitor.log* was used for monitor-related events
- *server-<id>.log* was used for each individual server

Logs are written to separate files. This makes it easier to trace system behaviours and debug issues during system failure

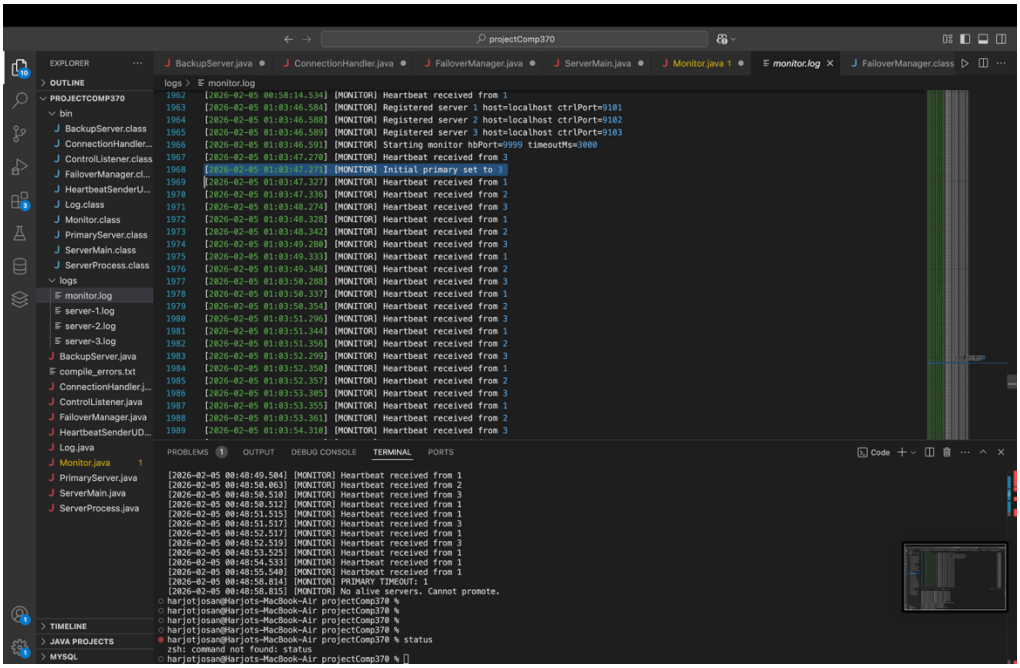
```
harjotjosan@Harjots-MacBook-Air projectComp370 % cd ~/Desktop/projectComp370
java -cp bin Monitor
[2026-02-05 00:49:44.689] [MONITOR] Registered server 1 host=localhost ctrlPort=
9101
[2026-02-05 00:49:44.694] [MONITOR] Registered server 2 host=localhost ctrlPort=
9102
[2026-02-05 00:49:44.694] [MONITOR] Registered server 3 host=localhost ctrlPort=
9103
[2026-02-05 00:49:44.697] [MONITOR] Starting monitor hbPort=9999 timeoutMs=3000
monitor>
```

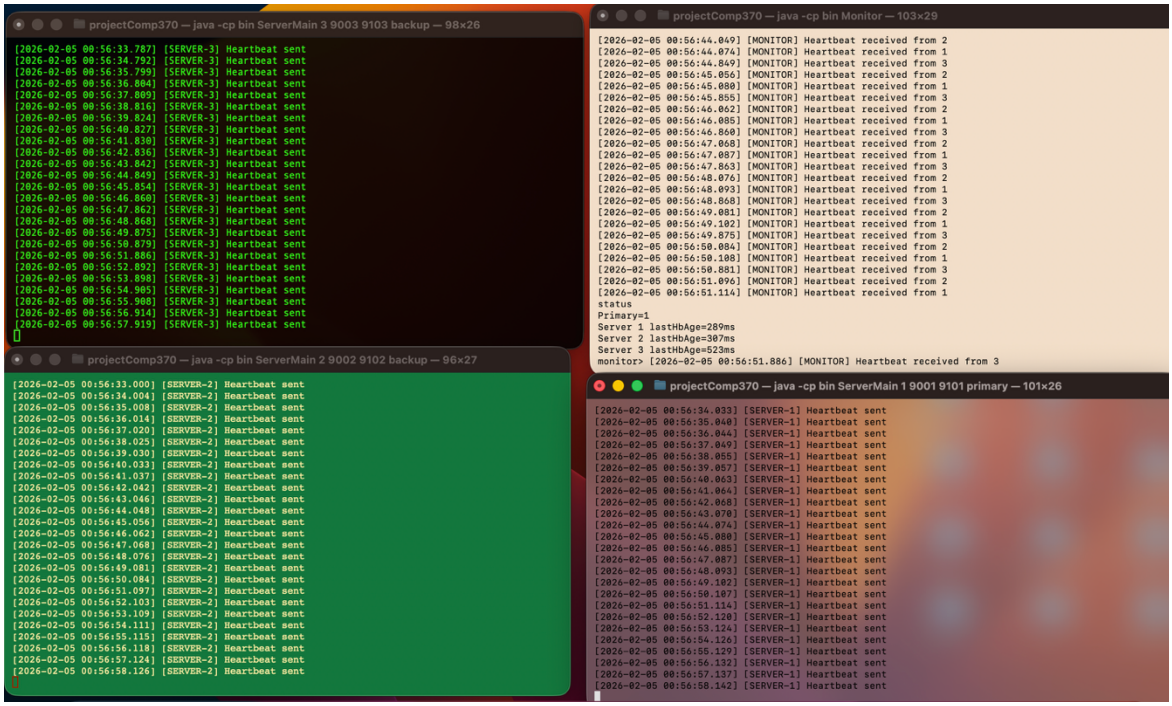
Timestamp format: [yyyy-mm-dd hh:mm:ss.sss]

7.2 Monitoring Mechanism

For this part, the monitoring mechanism is completed using UDP heartbeat messages. We used 3 servers and one monitor to test this. Each server sent heartbeat packets to be monitored at a fixed interval. The servers would say “[yyyy-mm-dd hh:mm:ss.sss] [server#] [heartbeat sent]” and the monitor would say “heartbeat received from server#.” The monitor recorded all the timestamps for the most recent heartbeat that it received.

The purpose of the monitoring mechanism is to see if a server has failed or not. If no heartbeat is received from a server, it would be considered failed. This mechanism lets the monitor to assess server health in real time with real-time updates. The monitored information is displayed in command line/terminal as well as in the log files.





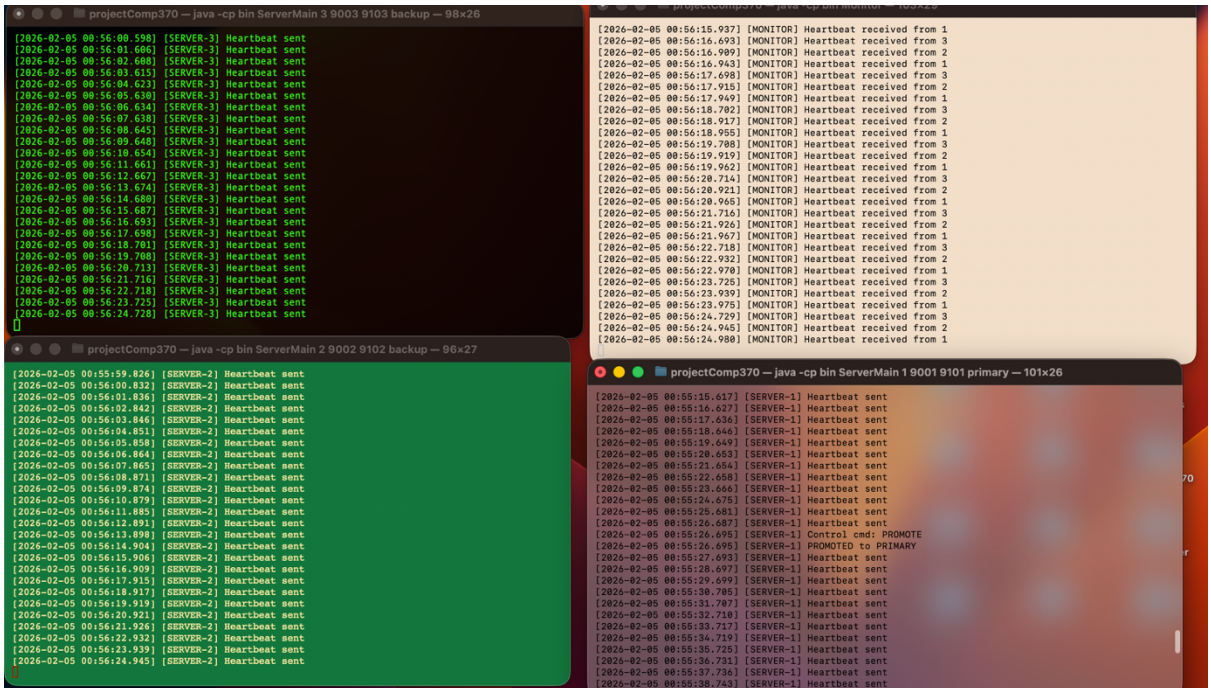
7.4 Admin Interface

A command-line administrative interface is given through the monitor. The interface would allow an admin to interact with the system while it is running. They can use commands like:

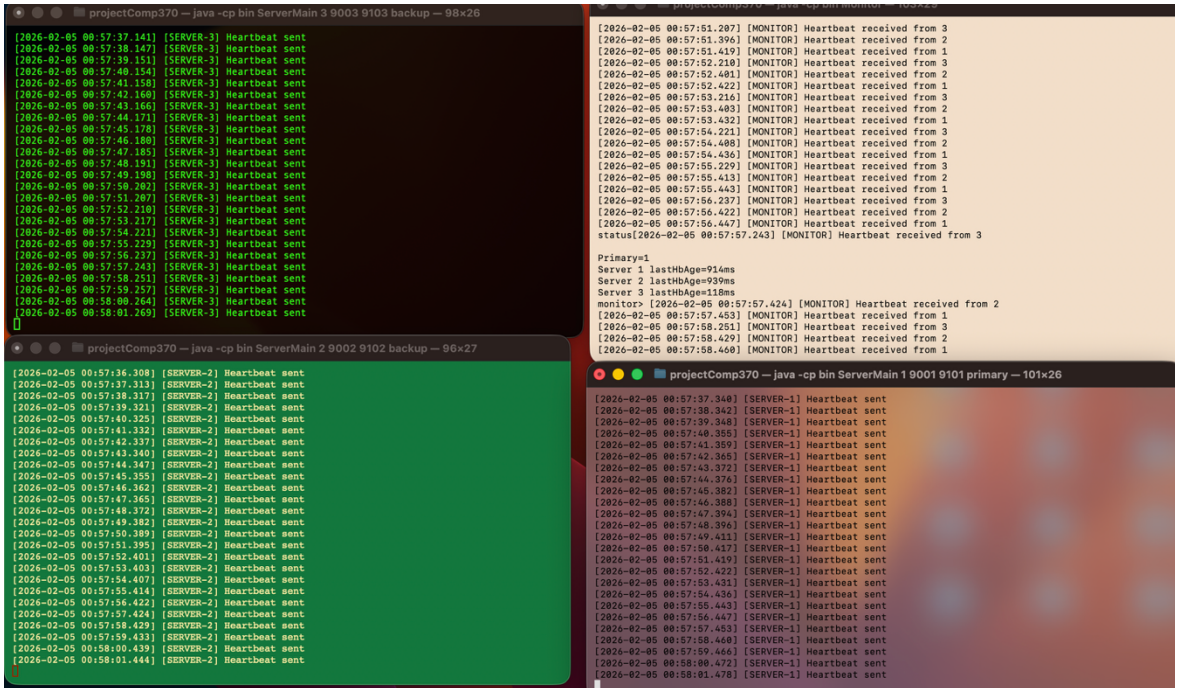
- status – Displays the health and heartbeat status of all servers
- primary – Displays the current primary server
- failover – Manually triggers a failover to promote a backup server

- Control + C – Shuts down the monitor or server

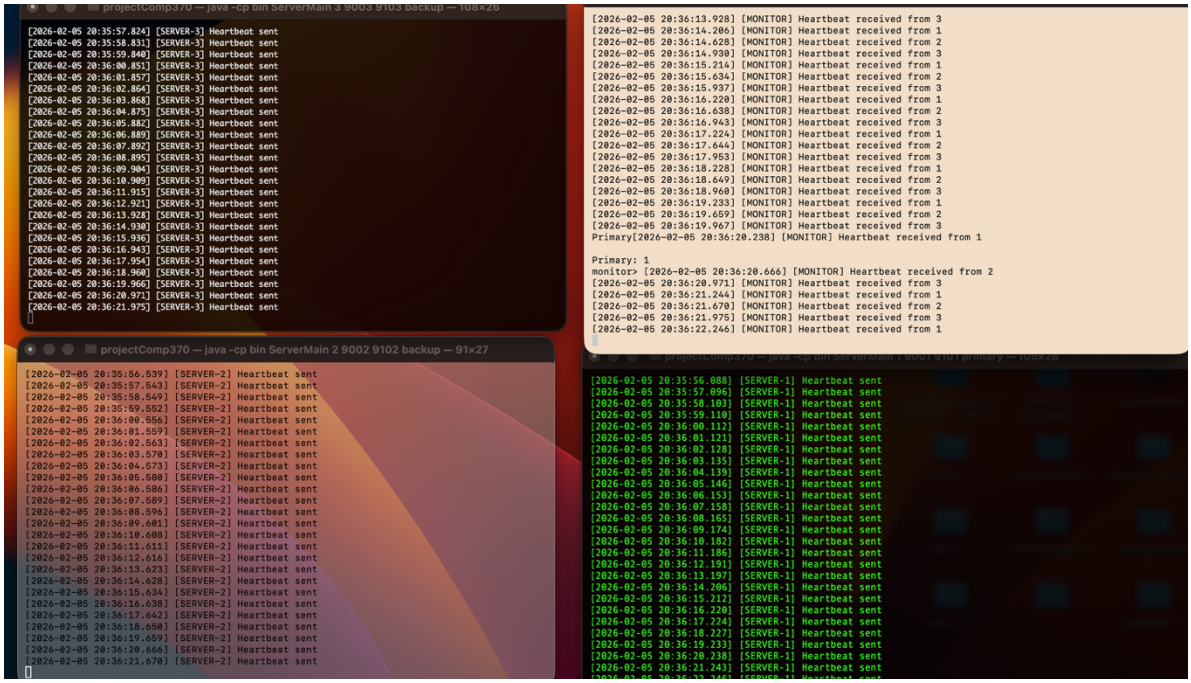
These commands can be used while the system is running. The interface enables runtime inspection and control without having to stop the system, which is important for testing and debugging fault-tolerant behaviours.



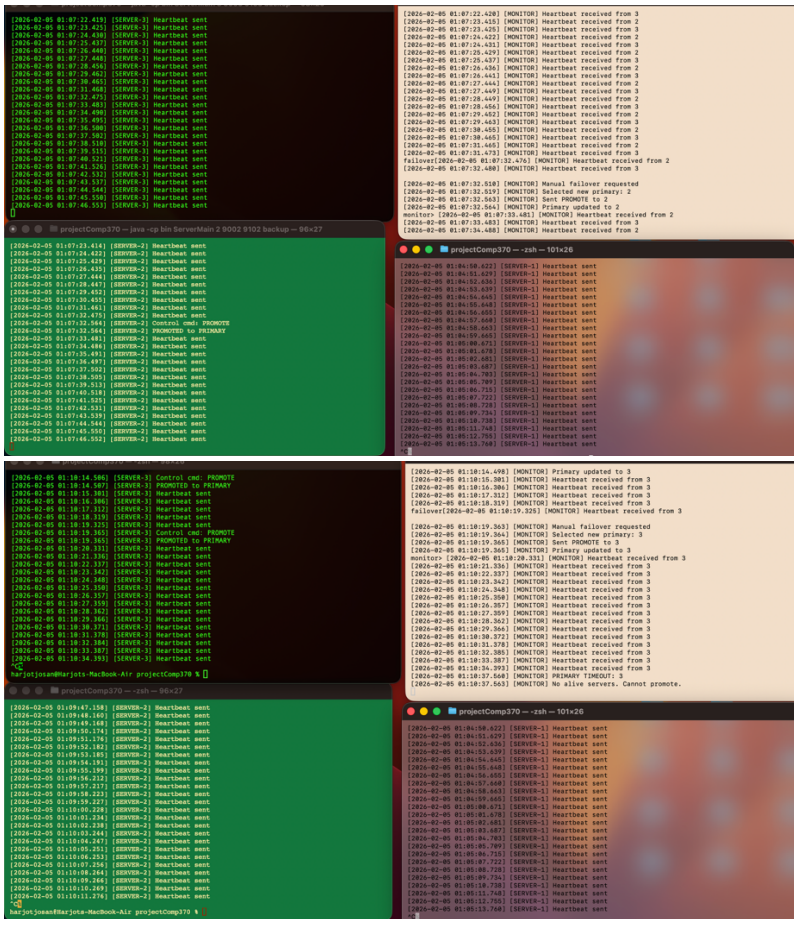
3 servers running and sending live updates to monitor (top right corner)



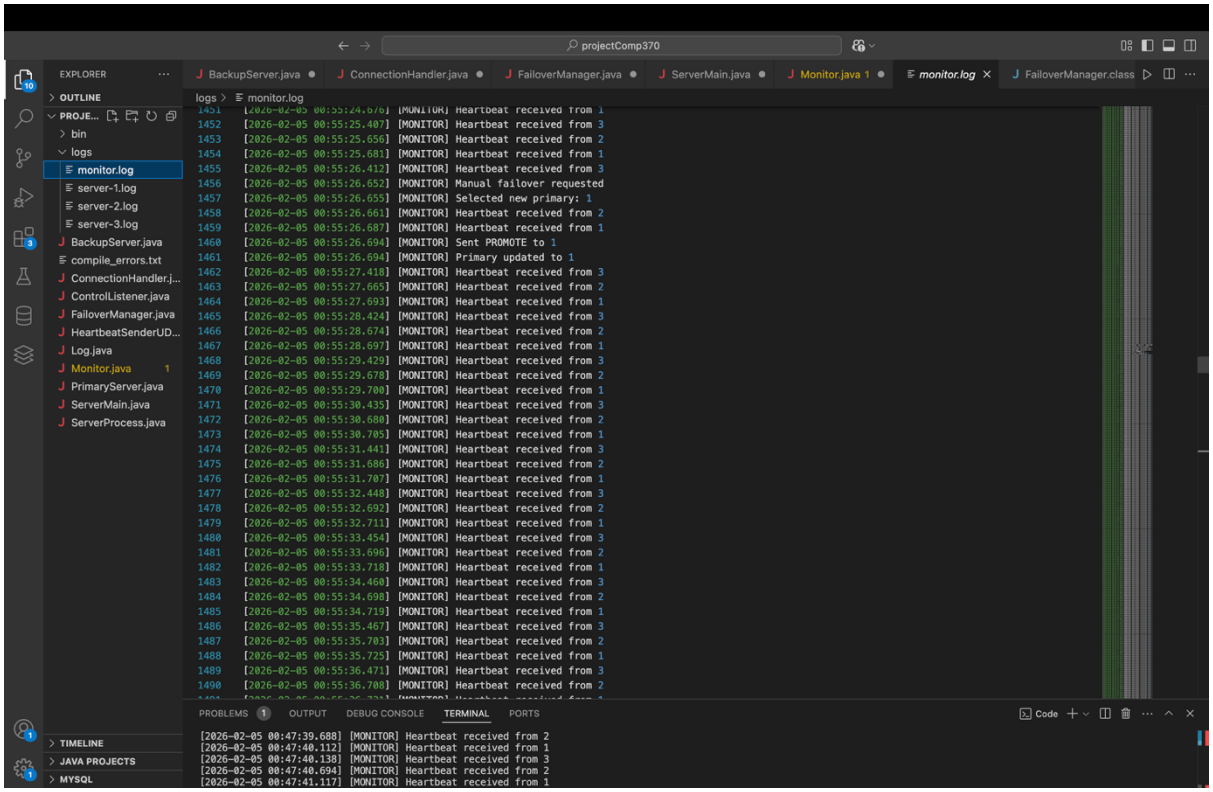
Using comand “Status”



Using comand “Primary”



Using command “Failover”



Live updates are stored being stored in “monitor.log” as well

Step 7 displayed success as it correctly implemented comprehensive logging, real-time monitoring, and an admin interface. Commands like “status, failover, and primary” were working correctly and giving live updates. Therefore, the system is observable, debuggable, and manageable during execution and testing.

Step 8: Testing and Failure Scenarios

8. Overview

Step 8 will validate the robustness and fault tolerance of SRMS under various failure conditions. We designed and executed several controlled test scenarios to observe system behaviour during failures, failure events, and network delays. We logged all these event timestamps for accurate results. These tests were conducted using three independent server instances and a single monitor, each running in its own JVM process.

8.1 Test Environment

- Operating System: macOS

- Language: Java
- Architecture: Primary–Backup server model
- Number of servers: 3 (1 primary, 2 backups)
- Heartbeat interval: 1 second
- Failure detection timeout: 3 seconds
- Communication: TCP for client/control, UDP for heartbeats

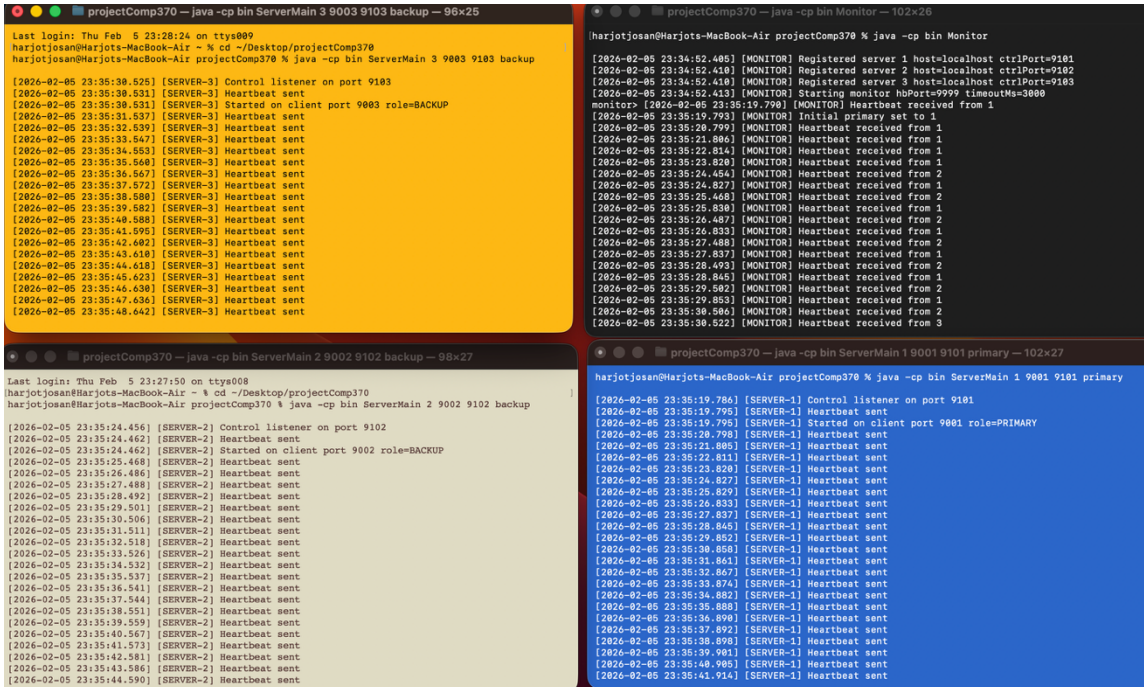
Scenario 1: Normal Operation

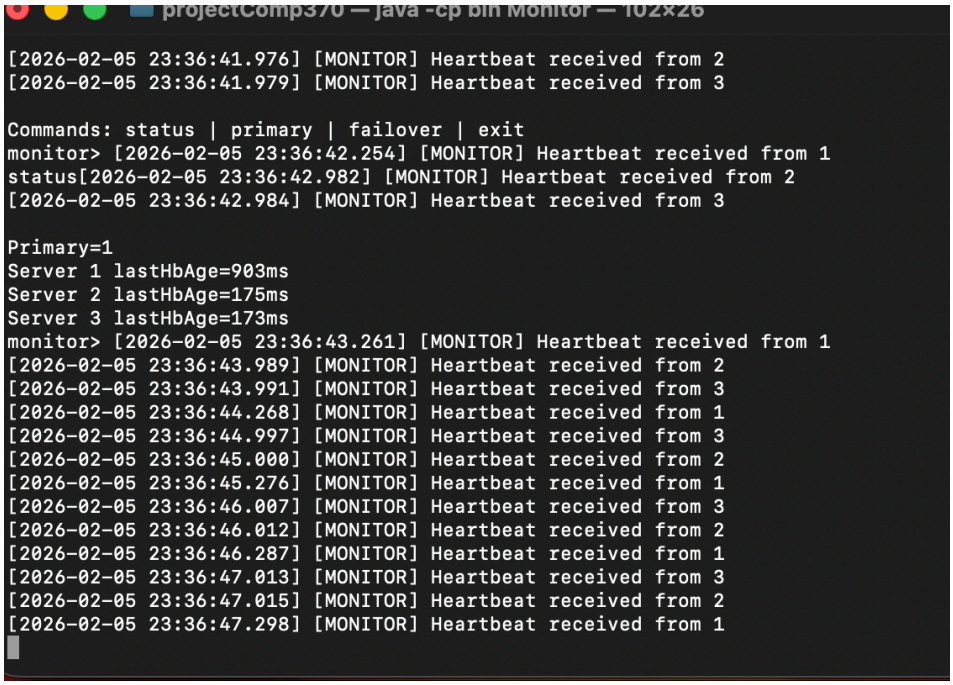
We will start monitor and 3 servers (1 primary and 2 backup) that send heartbeats to the monitor.

Expected Behaviour	**Observed Behaviour**
• A single primary server is selected • Monitor continuously receives heartbeats • No failover occurs • System remains stable	• Monitor successfully received heartbeats from all three servers • The primary server was correctly identified • No errors or failover events occurred

Recorded Metrics

- **Time of event:** [2026-02-05 23:35:19.793] [MONITOR] Initial primary set to 1
- **Detection time:** N/A
- **Failover time:** N/A
- **State consistency:** Consistent – all servers synchronized





Scenario 2: Primary Crash

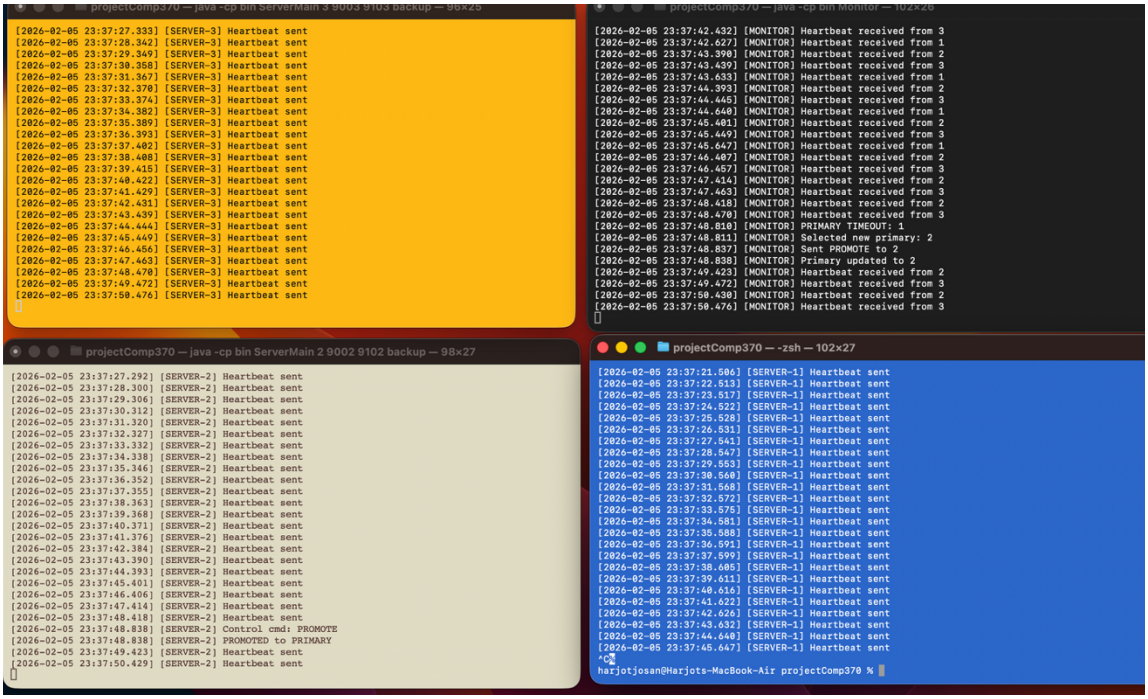
We will terminate primary server using control + c to stimulate a crash.

Expected Behaviour	**Observed Behaviour**
• Monitor stops detecting primary server heartbeats • Backup server is promoted to primary	• Primary servers' heartbeats are no longer sent to the monitor • A backup server was selected and promoted

• System continues operating with a new primary	• Promotion command was successfully received and acknowledged

Recorded Metrics

- **Time of event:** [2026-02-05 23:37:45.647] [SERVER-1] Heartbeat sent
- **Detection time:** [2026-02-05 23:37:48.810] [MONITOR] PRIMARY TIMEOUT: 1
  - 3.163 seconds
- **Failover time:** 0.028 seconds
  - Failure detection: [2026-02-05 23:37:48.810]
  - New primary server: [2026-02-05 23:37:48.838] [SERVER-2] PROMOTED to PRIMARY
- **State consistency:** Consistent – backup promoted cleanly



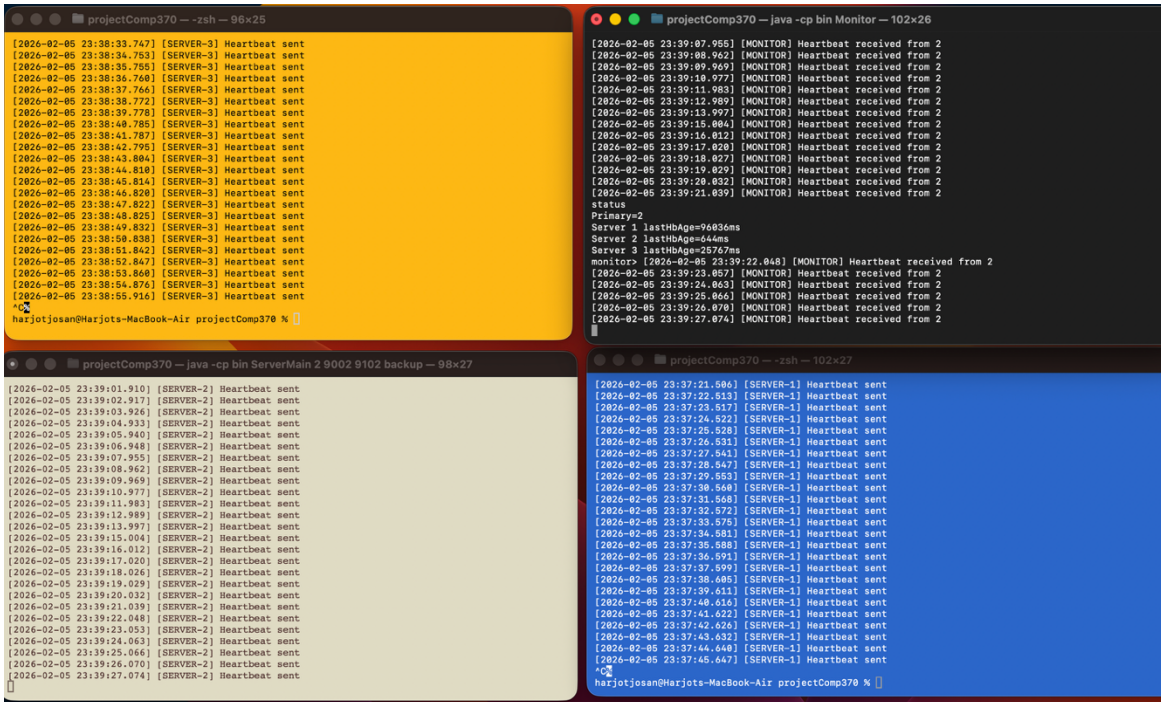
Scenario 3: Backup Crash

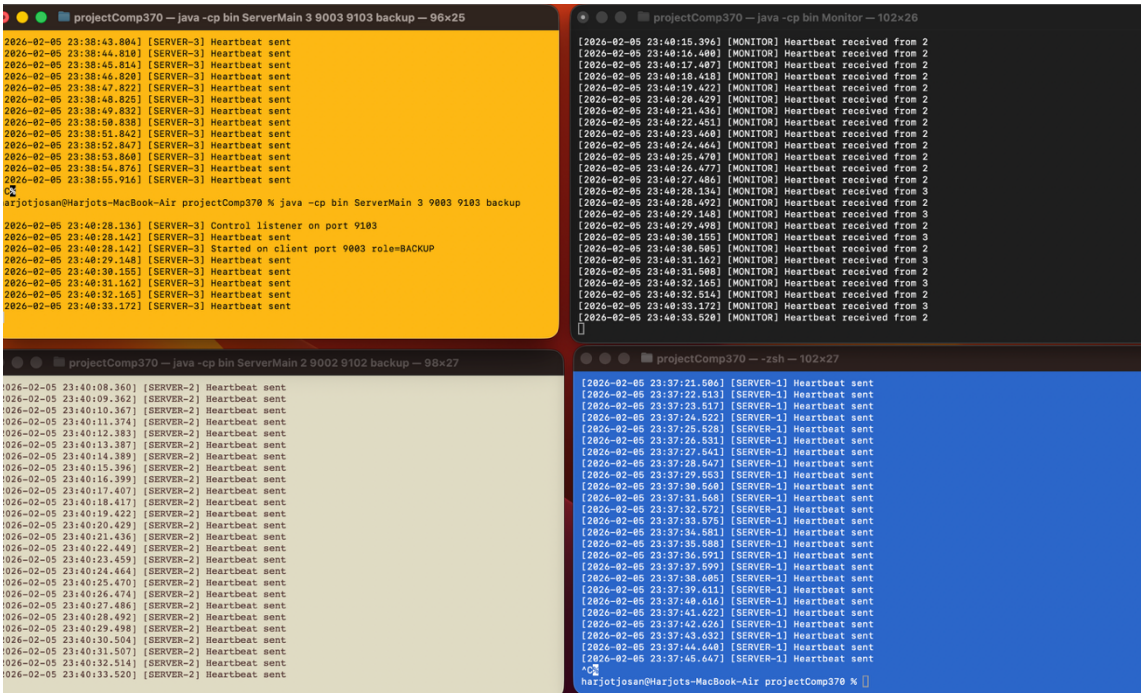
We will terminate the backup server, and the primary server will continue to operate.

Expected Behaviour	Observed Behaviour
• Primary continues sending heartbeats • No failover occurs • Monitor logs the missing heartbeat	• Monitor detected loss of heartbeats from the backup server • Primary server continued operating normally • No promotion or role change occurred • The backup server resumed sending heartbeats after restarting it

Recorded Metrics

- **Time of event:** [2026-02-05 23:38:55.916] [SERVER-3] Heartbeat sent
- **Detection time:** Immediate (heartbeat absence noted)
- **Failover time:** N/A
- **State consistency:** System state remained consistent and the monitor didn't tell us either that Backup timed out so we restarted backup and it rejoined as a backup





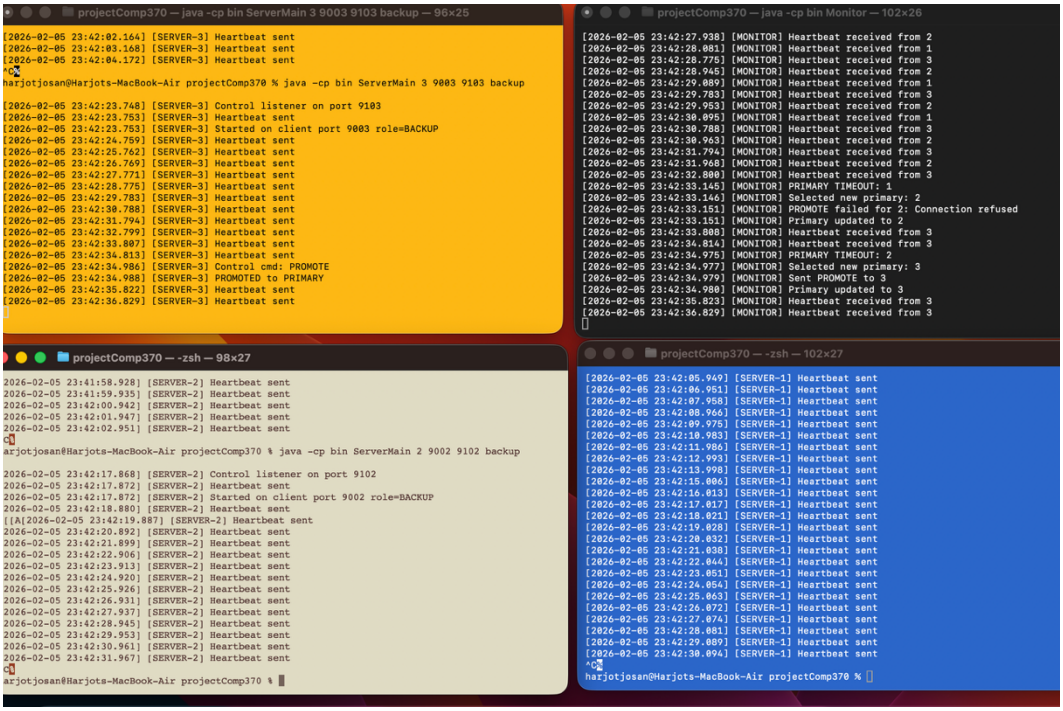
Scenario 4: Simultaneous Failures

The primary server and one backup server are terminated at the same time.

Expected Behaviour	**Observed Behaviour**
• Remaining server is promoted to primary server	• Monitor detected loss of heartbeats from both failed servers • Remaining server is successfully promoted to primary • System continued operation with a single server

Recorded Metrics

- **Time of event:**
  - [2026-02-05 23:41:35.866] [SERVER-2] Heartbeat sent
  - [2026-02-05 23:41:36.584] [SERVER-3] Heartbeat sent
- **Detection time:** ≈3.5 seconds
  - [2026-02-05 23:41:39.747] [MONITOR] No alive servers. Cannot promote.
- **Failover time:** <1 second
- **State consistency:** No split-brain condition occurred



Scenario 5: Network Delay Simulation

Artificial delays are introduced in heartbeat transmission by adding sleep delays to the heartbeat sender logic.

Expected Behaviour	**Observed Behaviour**
• Monitor tolerates small delays	• Minor delays did not trigger failover

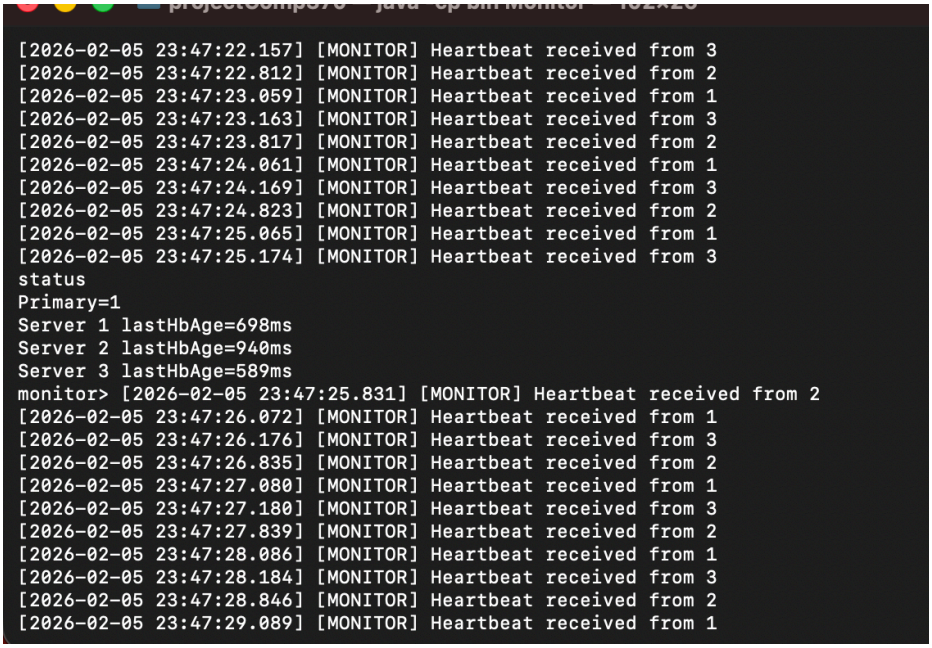
• No false failover occurs unless delays exceed timeout threshold	• Failover occurred only when heartbeat delays exceeded the configured timeout

Recorded Metrics

- **Time of event:** [2026-02-05 23:47:25.831] [MONITOR] Heartbeat received from 2
- **Detection time:** 1 minute 7.370 seconds
  - [2026-02-05 23:46:18.461] – last heartbeat received
  - [2026-02-05 23:47:25.831]- first heartbeat received again
- **Failover time:** Consistent with timeout configuration
- **State consistency:** No inconsistencies observed

```
        Log.info("SERVER-" + serverId, msg: "Heartbeat sent");

        Thread.sleep(intervalMs + 1500);
    }
  catch (Exception e) {
    Log.info("SERVER-" + serverId, "Heartbeat sender stopped: " + e.getMessage());
```



Scenario 6: Recovery

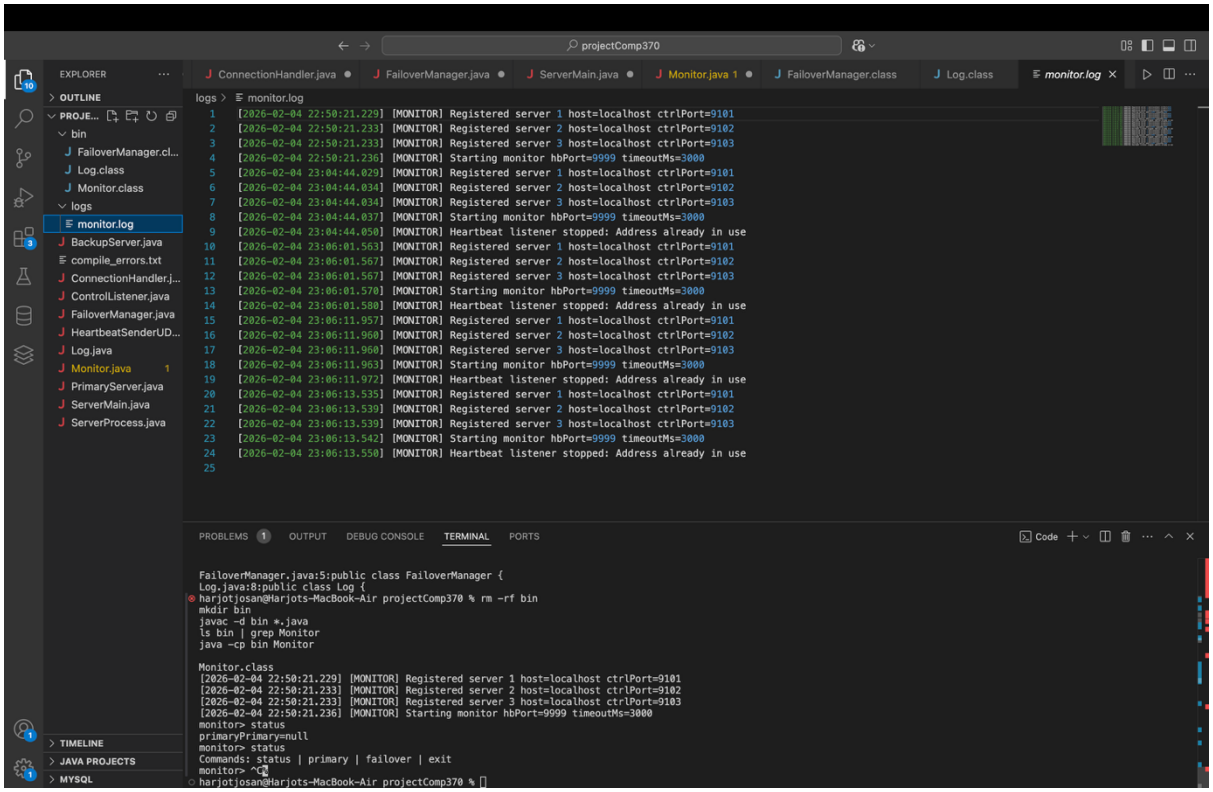
All crashed servers are restarted and allowed to rejoin the cluster.

Expected Behaviour	**Observed Behaviour**
• Restarted servers send heartbeats • Monitor recognizes them as backups • No unnecessary role changes occur	• Restarted servers successfully rejoined • Monitor logged renewed heartbeat reception • Servers resumed backup roles correctly

Recorded Metrics

- **Time of event:** [2026-02-05 23:47:15.550] [MONITOR] Selected new primary: 1
- **Detection time:** Immediate upon heartbeat reception
- **Failover time:** N/A
- **State consistency:** System returned to stable state

026-02-05 23:48:06.417] [SERVER-3] Heartbeat sent
026-02-05 23:48:07.420] [SERVER-3] Heartbeat sent
026-02-05 23:48:08.425] [SERVER-3] Heartbeat sent
026-02-05 23:48:09.431] [SERVER-3] Heartbeat sent
026-02-05 23:48:10.436] [SERVER-3] Heartbeat sent
026-02-05 23:48:11.444] [SERVER-3] Heartbeat sent
026-02-05 23:48:12.448] [SERVER-3] Heartbeat sent
026-02-05 23:48:13.455] [SERVER-3] Heartbeat sent
026-02-05 23:48:14.462] [SERVER-3] Heartbeat sent
026-02-05 23:48:15.469] [SERVER-3] Heartbeat sent
026-02-05 23:48:16.476] [SERVER-3] Heartbeat sent
026-02-05 23:48:17.484] [SERVER-3] Heartbeat sent
026-02-05 23:48:18.487] [SERVER-3] Heartbeat sent
026-02-05 23:48:19.495] [SERVER-3] Heartbeat sent
026-02-05 23:48:20.503] [SERVER-3] Heartbeat sent
026-02-05 23:48:21.511] [SERVER-3] Heartbeat sent
026-02-05 23:48:22.518] [SERVER-3] Heartbeat sent
026-02-05 23:48:23.521] [SERVER-3] Heartbeat sent
026-02-05 23:48:24.524] [SERVER-3] Heartbeat sent
026-02-05 23:48:25.528] [SERVER-3] Heartbeat sent
026-02-05 23:48:26.535] [SERVER-3] Heartbeat sent
026-02-05 23:48:27.542] [SERVER-3] Heartbeat sent
026-02-05 23:48:28.546] [SERVER-3] Heartbeat sent
026-02-05 23:48:29.550] [SERVER-3] Heartbeat sent
[2026-02-05 23:48:14.131] [MONITOR] Heartbeat received from 2
[2026-02-05 23:48:14.462] [MONITOR] Heartbeat received from 3
[2026-02-05 23:48:14.501] [MONITOR] Heartbeat received from 1
[2026-02-05 23:48:15.138] [MONITOR] Heartbeat received from 2
[2026-02-05 23:48:15.469] [MONITOR] Heartbeat received from 3
[2026-02-05 23:48:15.508] [MONITOR] Heartbeat received from 1
Commands: status | primary | failover | exit
monitor> [2026-02-05 23:48:16.142] [MONITOR] Heartbeat received from 2
[2026-02-05 23:48:16.477] [MONITOR] Heartbeat received from 3
[2026-02-05 23:48:16.512] [MONITOR] Heartbeat received from 1
[2026-02-05 23:48:17.149] [MONITOR] Heartbeat received from 2
[2026-02-05 23:48:17.484] [MONITOR] Heartbeat received from 3
[2026-02-05 23:48:17.515] [MONITOR] Heartbeat received from 1
[2026-02-05 23:48:18.155] [MONITOR] Heartbeat received from 2
[2026-02-05 23:48:18.488] [MONITOR] Heartbeat received from 3
[2026-02-05 23:48:18.522] [MONITOR] Heartbeat received from 1
status
Primary=2
Server 1 lastHbAge=412ms
Server 2 lastHbAge=780ms
Server 3 lastHbAge=446ms
monitor> [2026-02-05 23:48:19.160] [MONITOR] Heartbeat received from 2
[2026-02-05 23:48:19.495] [MONITOR] Heartbeat received from 3
[2026-02-05 23:48:19.526] [MONITOR] Heartbeat received from 1
[2026-02-05 23:48:20.164] [MONITOR] Heartbeat received from 2
projectComp370 — java -cp bin ServerMain 2 9002 9102 backup — 98×27
26-02-05 23:48:04.062] [SERVER-2] Heartbeat sent
26-02-05 23:48:05.071] [SERVER-2] Heartbeat sent
26-02-05 23:48:06.085] [SERVER-2] Heartbeat sent
26-02-05 23:48:07.093] [SERVER-2] Heartbeat sent
26-02-05 23:48:08.095] [SERVER-2] Heartbeat sent
26-02-05 23:48:09.101] [SERVER-2] Heartbeat sent
26-02-05 23:48:10.109] [SERVER-2] Heartbeat sent
26-02-05 23:48:11.116] [SERVER-2] Heartbeat sent
26-02-05 23:48:12.122] [SERVER-2] Heartbeat sent
26-02-05 23:48:13.126] [SERVER-2] Heartbeat sent
26-02-05 23:48:14.131] [SERVER-2] Heartbeat sent
26-02-05 23:48:15.137] [SERVER-2] Heartbeat sent
26-02-05 23:48:16.142] [SERVER-2] Heartbeat sent
26-02-05 23:48:17.149] [SERVER-2] Heartbeat sent
26-02-05 23:48:18.154] [SERVER-2] Heartbeat sent
26-02-05 23:48:19.159] [SERVER-2] Heartbeat sent
26-02-05 23:48:20.163] [SERVER-2] Heartbeat sent
26-02-05 23:48:21.171] [SERVER-2] Heartbeat sent
26-02-05 23:48:22.176] [SERVER-2] Heartbeat sent
26-02-05 23:48:23.177] [SERVER-2] Heartbeat sent
26-02-05 23:48:24.184] [SERVER-2] Heartbeat sent
26-02-05 23:48:25.192] [SERVER-2] Heartbeat sent
26-02-05 23:48:26.200] [SERVER-2] Heartbeat sent
26-02-05 23:48:27.206] [SERVER-2] Heartbeat sent
26-02-05 23:48:28.210] [SERVER-2] Heartbeat sent
26-02-05 23:48:29.213] [SERVER-2] Heartbeat sent
projectComp370 — java -cp bin ServerMain 1 9001 9101 backup — 102×27
[2026-02-05 23:47:45.181] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:46.185] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:47.190] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:48.192] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:49.199] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:50.206] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:51.212] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:52.218] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:53.222] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:54.224] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:55.232] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:56.240] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:57.243] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:58.250] [SERVER-1] Heartbeat sent
[2026-02-05 23:47:59.257] [SERVER-1] Heartbeat sent
^C
harjotjosan@Harjots-MacBook-Air projectComp370 % java -cp bin ServerMain 1 9001 9101 backup
[2026-02-05 23:48:09.466] [SERVER-1] Control listener on port 9101
[2026-02-05 23:48:09.471] [SERVER-1] Heartbeat sent
[2026-02-05 23:48:09.471] [SERVER-1] Started on client port 9001 role=BACKUP
[2026-02-05 23:48:10.477] [SERVER-1] Heartbeat sent
[2026-02-05 23:48:11.484] [SERVER-1] Heartbeat sent
[2026-02-05 23:48:12.491] [SERVER-1] Heartbeat sent
[2026-02-05 23:48:13.495] [SERVER-1] Heartbeat sent
[2026-02-05 23:48:14.501] [SERVER-1] Heartbeat sent
[2026-02-05 23:48:15.508] [SERVER-1] Heartbeat sent


projectComp370
EXPLORER
OUTLINE
PROJE...
bin
FailoverManager.cl...
Log.class
Monitor.class
logs
monitor.log
BackupServer.java
compile_errors.txt
ConnectionHandler.j...
ControlListener.java
FailoverManager.java
HeartbeatSenderUD...
Log.java
Monitor.java 1
PrimaryServer.java
ServerMain.java
ServerProcess.java
TIMELINE
JAVA PROJECTS
MYSQL
ConnectionHandler.java
FailoverManager.java
ServerMain.java
Monitor.java 1
FailoverManager.class
Log.class
monitor.log
logs > monitor.log
1 [2026-02-04 22:50:21.229] [MONITOR] Registered server 1 host=localhost ctrlPort=9101
2 [2026-02-04 22:50:21.233] [MONITOR] Registered server 2 host=localhost ctrlPort=9102
3 [2026-02-04 22:50:21.233] [MONITOR] Registered server 3 host=localhost ctrlPort=9103
4 [2026-02-04 22:50:21.236] [MONITOR] Starting monitor hbPort=9999 timeoutMs=3000
5 [2026-02-04 23:04:44.029] [MONITOR] Registered server 1 host=localhost ctrlPort=9101
6 [2026-02-04 23:04:44.034] [MONITOR] Registered server 2 host=localhost ctrlPort=9102
7 [2026-02-04 23:04:44.034] [MONITOR] Registered server 3 host=localhost ctrlPort=9103
8 [2026-02-04 23:04:44.037] [MONITOR] Starting monitor hbPort=9999 timeoutMs=3000
9 [2026-02-04 23:04:44.050] [MONITOR] Heartbeat listener stopped: Address already in use
10 [2026-02-04 23:06:01.563] [MONITOR] Registered server 1 host=localhost ctrlPort=9101
11 [2026-02-04 23:06:01.567] [MONITOR] Registered server 2 host=localhost ctrlPort=9102
12 [2026-02-04 23:06:01.567] [MONITOR] Registered server 3 host=localhost ctrlPort=9103
13 [2026-02-04 23:06:01.570] [MONITOR] Starting monitor hbPort=9999 timeoutMs=3000
14 [2026-02-04 23:06:01.580] [MONITOR] Heartbeat listener stopped: Address already in use
15 [2026-02-04 23:06:11.957] [MONITOR] Registered server 1 host=localhost ctrlPort=9101
16 [2026-02-04 23:06:11.960] [MONITOR] Registered server 2 host=localhost ctrlPort=9102
17 [2026-02-04 23:06:11.960] [MONITOR] Registered server 3 host=localhost ctrlPort=9103
18 [2026-02-04 23:06:11.963] [MONITOR] Starting monitor hbPort=9999 timeoutMs=3000
19 [2026-02-04 23:06:11.972] [MONITOR] Heartbeat listener stopped: Address already in use
20 [2026-02-04 23:06:13.535] [MONITOR] Registered server 1 host=localhost ctrlPort=9101
21 [2026-02-04 23:06:13.539] [MONITOR] Registered server 2 host=localhost ctrlPort=9102
22 [2026-02-04 23:06:13.539] [MONITOR] Registered server 3 host=localhost ctrlPort=9103
23 [2026-02-04 23:06:13.542] [MONITOR] Starting monitor hbPort=9999 timeoutMs=3000
24 [2026-02-04 23:06:13.550] [MONITOR] Heartbeat listener stopped: Address already in use
25
PROBLEMS 1
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Code
FailoverManager.java:5:public class FailoverManager {
Log.java:8:public class Log {
harjotjosan@Harjots-MacBook-Air projectComp370 % rm -rf bin
mkdir bin
javac -d bin *.java
ls bin | grep Monitor
java -cp bin Monitor
Monitor.class
[2026-02-04 22:50:21.229] [MONITOR] Registered server 1 host=localhost ctrlPort=9101
[2026-02-04 22:50:21.233] [MONITOR] Registered server 2 host=localhost ctrlPort=9102
[2026-02-04 22:50:21.233] [MONITOR] Registered server 3 host=localhost ctrlPort=9103
[2026-02-04 22:50:21.236] [MONITOR] Starting monitor hbPort=9999 timeoutMs=3000
monitor> status
primaryPrimary=null
monitor> status
Commands: status | primary | failover | exit
monitor> ^C
harjotjosan@Harjots-MacBook-Air projectComp370 %